

Vote, or Search a Tree

DATA 443/643 · Week 13, note 13.2

Week 13

American University

Ahmad Mousavi

Ideas from Stanford CS224N 2026 L12 (SC +17.9 pp GSM8K). Original slides; that course is not copied.

Lecture plan

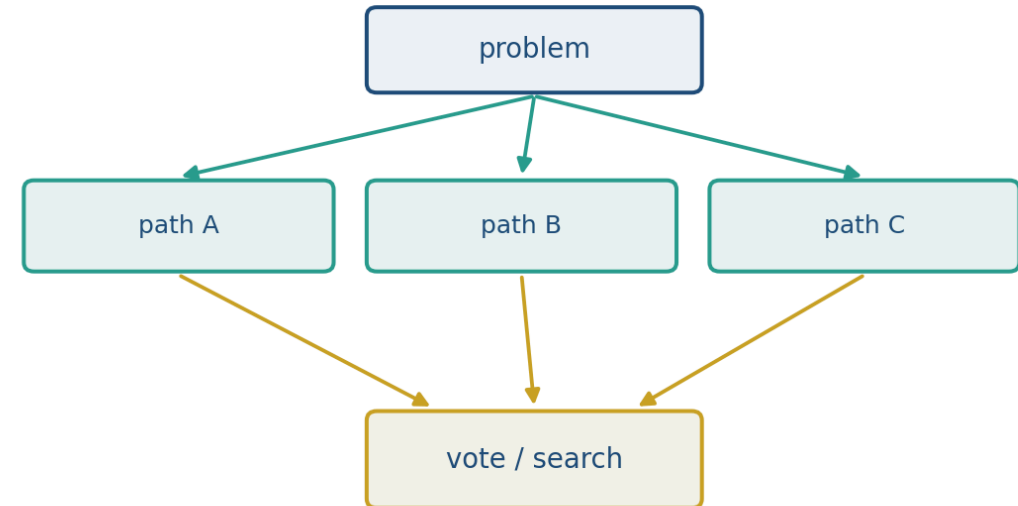
- 1 k traces, majority on the parsed answer
- 2 Wang et al.: +17.9 pp on GSM8K
- 3 ToT scores partial nodes
- 4 A shared bug still wins the vote

Key goal: vote on answers. $k=1$ is greedy CoT.

Test-time compute as k forwards

- Temperature > 0 . Parse each box. Majority.
- CS224N L12: extra tokens are extra compute. SC spends k of them.
- Do not assign DeepSeek-R1 / GRPO as homework.

Self-consistency votes; Tree-of-Thoughts searches partial steps



Majority on the parsed answer

$$\hat{a} = \operatorname{argmax}_a \quad | \{i: \text{parse}(y_i) = a\} |$$

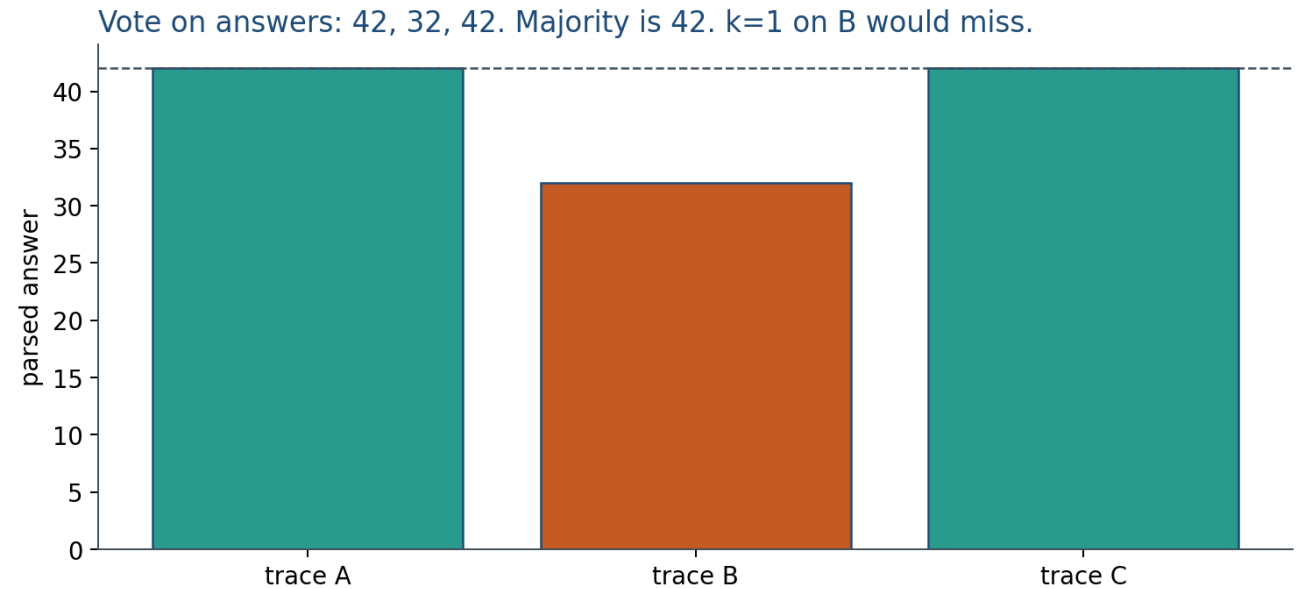
Vote on the number, not on the wording of the steps.

Tie-break is a method choice. Write it down.

Temperature 0 makes all k traces the same.

Lab 13 numbers, then GSM8K

- $23+19$: answers 42, 32, 42. Majority 42.
- $k = 1$ on B scores 32 and misses.
- Wang et al., cited in CS224N L12: +17.9 pp vs greedy CoT.



Two ways to spend the budget

Self-consistency

i.i.d. full traces. Vote. Shared formula bugs survive.

Tree-of-Thoughts

Expand partial thoughts, score, prune. ReAct (Week 14) calls a tool instead.

What to take from this lecture

Plot k

Accuracy vs k, plus greedy CoT at k=1.

GSM8K punch

+17.9 pp is a lift, not a proof of faithful steps.

Lab 13

Vote on constructed answers. Then flag 13.3.